Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 June 2026
Team ID	
Project Name	House Rent
Maximum Marks	4 Marks

Technical Architecture:

The House Rent Management System is designed using a **three-tier architecture** consisting of the **Presentation Layer (Frontend)**, **Application Layer (Backend)**, and **Data Layer (Database)**. The system allows Tenants, Landlords, and Administrators to interact with the application through a web interface. User requests are processed by the backend server, which communicates with the MongoDB database to store and retrieve data securely.

The architecture includes user authentication using **JWT (JSON Web Token)**, password encryption using **bcryptjs**, RESTful APIs developed with **Express.js**, and data storage in **MongoDB**. The frontend is built using **React.js (Vite)**, which communicates with the backend through HTTP requests.

Example: Order processing during pandemics for offline mode

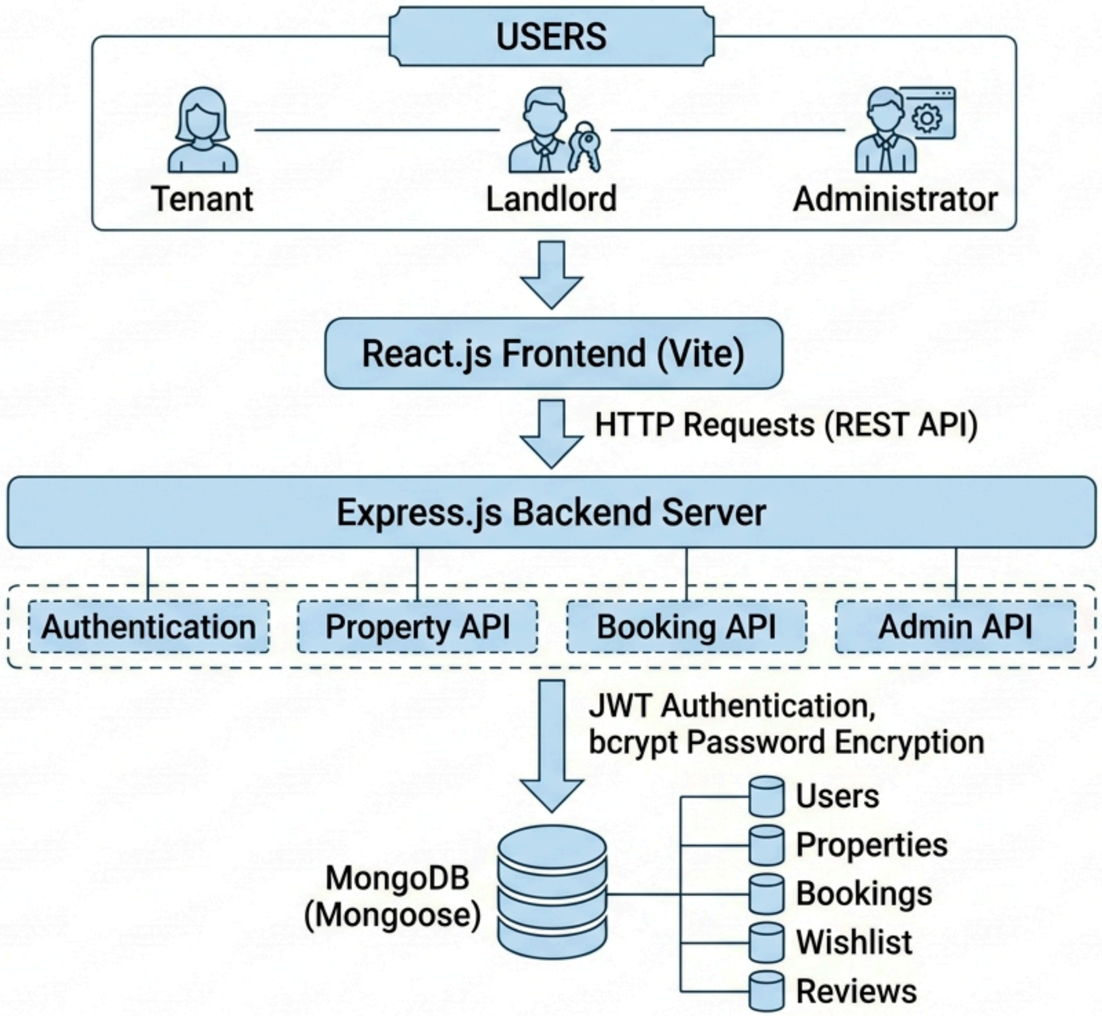
USERS
Tenant
Landlord
Administrator
React.js Frontend (Vite)
HTTP Requests (REST API)
Express.js Backend Server
Authentication
Property API
Booking API
Admin API
JWT Authentication,
bcrypt Password Encryption
MongoDB
(Mongoose)
Users
Properties
Bookings
Wishlist
Reviews

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web application interface for Tenant, Landlord, and Administrator	React.js (Vite), HTML5, CSS3, JavaScript
2	User Authentication	Authentication and freencrypt authentication	Express.js (JWT
3	Property Management	Properties for property management, and manioned server	Express.js, HTTP
4	Booking/Wishlist/ Reviews	Calendars for booking, stscipiants, and administrators	Whaatt.js, Wishlist, REST API
5	Cloud Database	Cloud-hosted MongoDB database (if deployed)	MongoDB Atlas
6	Cloud Database	Cloud-hosted MongoDB database (if deployed)	MongoDB Atlas
7	Cloud Deployment	Cloud-hosted MongoDB deployment	MongoDB Atlas
8	File Storage	File storage and foled folders	File Storage
9	External APIs	Internet crais connectiured with connectin, and gear APIs	External Gear API, Cloonat Cloud near, API
10	Machine Learning Model	N/A	N/A
11	Infrastructure (Server / Cloud)	Application deployment for development and production	Local: Node.js Server, MongoDB Local, Cloud: Vercel (Frontend), Render/Railway (Backend), Backend), MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frontend and backend development frameworks used in the application	React.js, Vite, Node.js, Express.js, MongoDB, Mongoose
2	Security Implementations Implementations	Secure authentication, password encryption, authorization, and protected APIs	JWT Authentication, bcryptjs Password Hashing, bcryptjs Password Hashing, CORS, Environment Variables (.env), OWASP Best Practices
3	Scalable Architecture	Three-tier architecture separating presentation, application, and database layers.	React.js + Express.js + MongoDB (3-Tier Architecture)
4	Availability	Application can be deployed on cloud servers with continuous availability and centralized database.	Vercel, Render/Railway, MongoDB Atlas
5	Performance	Optimized REST APIs, efficient MongoDB queries, lightweight frontend, asynchronous request handling.	Express.js, MongoDB Indexing, React Virtual DOM, HTTP/JSON APIs